CHAPTER 5

WEB DESIGNING USING HTML

CHAPTER ESSENCE AT A GLANCE:
Design accessible, standards-compliant web pages using HTML semantic tags, tables, nested lists, multi-input interactive forms, and Cascading Style Sheets.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Document Tree Hierarchy:** Every hypertext markup document follows a strict tree hierarchy with root elements containing head metadata for configuration and body containers for visible page content.

• **Container and Empty Tag Semantics:** Container elements enclose content between paired opening and closing markup tags, whereas empty standalone elements insert discrete standalone media or structural breaks directly into the document stream.

• **Tabular Layout and Cell Spanning:** Hypertext tables organize structured data into synchronized rows and cells, utilizing horizontal column spanning and vertical row spanning to merge adjacent coordinates.

• **Interactive Form Data Capture:** Web forms collect user inputs through specialized controls including text fields, radio buttons, checkboxes, and dropdown lists, transmitting bundled payloads via retrieval or post request methods.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\langle \text{html} \rangle \longrightarrow \langle \text{head} \rangle (\text{Metadata}) + \langle \text{body} \rangle (\text{Content})$$

Hypertext document structural tree hierarchy

$$\text{colspan} = N \ (\text{Columns Across}) \quad | \quad \text{rowspan} = M \ (\text{Rows Down})$$

HTML table cell merging coordinate dimensions

$$\text{Inline Style} > \text{Internal Style Sheet} > \text{External Style Sheet}$$

Cascading Style Sheets specificity and override priority

$$\text{Aspect Ratio} = \frac{\text{Rendered Image Width}}{\text{Rendered Image Height}}$$

Responsive web graphic scaling ratio in HTML

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Empty tags (such as br, hr, img, input) do not take closing tags, whereas container tags must always be cleanly closed.

• **Alert:** Radio buttons must share the exact same name attribute value to enforce mutual exclusivity.

• **Alert:** External stylesheets linked in the head section provide maximum maintainability over inline style attributes.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Explain the basic structure of an HTML document (<html>, <head>, <title>, <body>). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Structure):** <!DOCTYPE html>n<html>n <head>n <title>Page Title</title>n <meta charset='UTF-8'>n </head>n <body>n <h1>Welcome</h1>n <p>Body content visible to users goes here.</p>n </body>n</html> **[2 Marks]**

• **Key Exam Result:** <html> containing <head> (metadata) and <body> (visible content)

$$\langle \text{html} \rangle \longrightarrow \langle \text{head} \rangle (\text{Metadata}) + \langle \text{body} \rangle (\text{Content})$$

Hypertext document structural tree hierarchy

2. Differentiate between Container Tags and Empty Tags in HTML with examples. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Container Tags):** Tags that have both opening and closing tags enclosing text or other elements: <tag> ... </tag>. Examples: <html>...</html>, <p>...</p>, **....** **[1 Mark]**

• **Step 2 (Empty Tags):** Tags that do not have closing tags and contain no content: <tag> (or <tag/>). Examples:
 (line break), <hr> (horizontal rule), <img> (image). **[1 Mark]**

• **Key Exam Result:** Container: <p>...</p> | Empty:
, <hr>, <img>

$$\langle \text{html} \rangle \longrightarrow \langle \text{head} \rangle (\text{Metadata}) + \langle \text{body} \rangle (\text{Content})$$

Hypertext document structural tree hierarchy

3. Explain Unordered Lists (<ul>) and Ordered Lists (<ol>) in HTML with their attributes. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Unordered List <ul>):** Bulleted list with type attribute ('disc', 'circle', 'square'): Physics
Chemistry. **[1.5 Marks]**

• **Step 2 (Ordered List <ol>):** Numbered list with type attribute ('1', 'A', 'a', 'I', 'i') and start attribute: <ol type='A' start='1'>
Chapter 1
Chapter 2. **[1.5 Marks]**

• **Key Exam Result:** for bullets; for numbers with type and start attributes

$$\langle \text{html} \rangle \longrightarrow \langle \text{head} \rangle (\text{Metadata}) + \langle \text{body} \rangle (\text{Content})$$

Hypertext document structural tree hierarchy

4. Explain HTML tables: <table>, <tr>, <th>, <td> tags and attributes (border, cellpadding, rowspan, colspan). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Table Tags):** <table> defines table, <tr> defines table row, <th> defines header cell (bold, centered), <td> defines standard data cell. **[1.5 Marks]**

• **Step 2 (Attributes):** border: thickness of table border (border='1'). cellspacing: space between adjacent cells. cellpadding: space between cell border and cell content. rowspan or colspan: merges rows or columns across cells. **[1.5 Marks]**

• **Key Exam Result:** <table>, <tr>, <th>, <td>; border, cellpadding, rowspan, colspan

$$\text{colspan} = N \ (\text{Columns Across}) \quad | \quad \text{rowspan} = M \ (\text{Rows Down})$$

HTML table cell merging coordinate dimensions

5. Explain hyperlinks (<a>) and image embedding (<img>) in HTML with their essential attributes. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Anchor Tag <a>):** Creates hyperlinks: Link Text. href specifies destination address; target='blank' opens in new tab. **[1.5 Marks]**

• **Step 2 (Image Tag <img>):** Embeds images: <img src='photo.jpg' alt='Student Photo' width='200' height='250'>. src: image path; alt: alternate text if image fails to load. **[1.5 Marks]**

• **Key Exam Result:** <a href='...'> (links) and <img src='...' alt='...'> (images)

$$\text{Aspect Ratio} = \frac{\text{Rendered Image Width}}{\text{Rendered Image Height}}$$

Responsive web graphic scaling ratio in HTML

6. Explain HTML Forms: <form> tag and various form controls (text, password, radio, checkbox, select, submit). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Form Tag):** <form action='submit.php' method='POST'>. action specifies server script URL; method specifies GET or POST. **[1 Mark]**

• **Step 2 (Controls):** Text box: <input type='text' name='uname'>. Password: <input type='password' name='pass'>. Radio button: <input type='radio' name='gender' value='M'>. Checkbox: <input type='checkbox' name='sub' value='CS'>. Dropdown list: <select name='state'><option>Kerala</option></select>. Submit button: <input type='submit' value='Submit'>. **[2 Marks]**

• **Key Exam Result:** <form> with text, password, radio, checkbox, select, submit

$$\text{GET: URL Query String (Visible)} \quad \longleftrightarrow \quad \text{POST: Body Payload (Secure)}$$

HTTP form transmission method comparison

7. Differentiate between GET and POST methods in HTML form data submission. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (GET Method):** Form data appended directly to URL (?name=value); visible in browser address bar; length limited (~2048 chars); can be bookmarked; insecure for passwords. **[1.5 Marks]**

• **Step 2 (POST Method):** Form data sent inside HTTP request message body; invisible in URL; unlimited data size; supports binary file uploads; secure for confidential data. **[1.5 Marks]**

• **Key Exam Result:** GET (data in URL, limited, visible) vs POST (data in body, unlimited, secure)

$$\text{GET: URL Query String (Visible)} \quad \longleftrightarrow \quad \text{POST: Body Payload (Secure)}$$

HTTP form transmission method comparison